Ministry of Electronics and Information Technology
Government of India

NIELIT Virtual Academy

INTERSHIP PROSPECTUS

Name of the Internship: Online Internship in Artificial Intelligence and Machine Learning using Python

Internship Code: IN31

Mode of Conduction: Online

Starting Date: 2nd June 2025

Last date of registration: 1st June 2025

Duration: 6 Weeks

Objective of the Course

- ❖ Provide basic theoretical concepts underlying machine learning.
- ❖ Provide a comprehensive understanding of artificial and machine learning concepts.
- ❖ Provide deep understanding with an effective algorithmic Perspective of AI & ML concepts.
- ❖ Provide hands-on to solve real-life AI & ML problems.

Outcome of the Course

- ❖ Provide the participants with in-depth knowledge AI-ML and its implementation

Course Fee: **Rs: 1000/-(inclusive of GST)**

Eligibility: **Pursuing an Undergraduate level course or above**

Methodology

- ✓ Teaching Mode: Self-Pace
- ✓ Access from anywhere anytime
- ✓ Content Access through e-learning portal
- ✓ Doubt Removal Session
- ✓ Covers both Theory & Practical
- ✓ Certification: On completion of the Mini Project

Registration Link: http://nva.nielit.gov.in

Contact Details:

- ✓ Course coordinator Name: Ms. Maithreyees S
- ✓ Email: trng.chennai@nielit.gov.in; ds@nielitchennai.edu.in
- ✓ Mobile number: **7598730125**

Course Structure:

Module No	Module Title
1	Introduction to Machine Learning & Data pre-processing
2	Machine Learning I
3	Machine Learning II
4	Artificial Neural Network
5	Deep Learning
6	Mini Project

Syllabus:

Detailed Syllabus

Module 1. Introduction to Machine Learning & Data pre-processing

- ❖ Objectives, Applications and types of Machine Learning
- ❖ Python libraries suitable for Machine Learning
- ❖ Dimensionality reduction-Principal Components Analysis
- ❖ Practical Implementation using Python
- ❖ Assignment-1

Module 2. Machine Learning I

- ❖ Building linear regression& KNN classifier
- ❖ Support Vector Machines
- ❖ Decision Tree
- ❖ Practical Implementation using Python
- ❖ Assignment-2

Module 3. Machine Learning II

- Unsupervised Learning – Introduction, K-Means Algorithm
- Fuzzy clustering
- Density Based Clustering
- Practical Implementation using Python
- Assignment-3

Module 4. Artificial Neural Network

- Introduction- Building Neural Network, Activation functions and Loss functions
- Training a single perceptron
- Back Propagation Algorithm
- Practical Implementation using Python
- Assignment - 4

Module 5. Deep Learning

- Introduction, CNN with Deep learning
- RNN- Architecture and Training
- LSTM and its application
- Practical Implementation using Python
- Assignment-5

Module 6. Project

- Capstone project
- Project Report Submission